

Title: Network Connectivity Troubleshooting and Command-Line Diagnostics

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1. Executive Summary & Objective

One of the most frequent tickets a Helpdesk Technician receives is a user stating they have lost connection to the network or cannot reach internal server. This document details how to diagnose a broken network adapter from the Windows command prompt, trace the root cause of a connection drop, and restore proper communications using standard network utilities.

2. Environmental Specifications

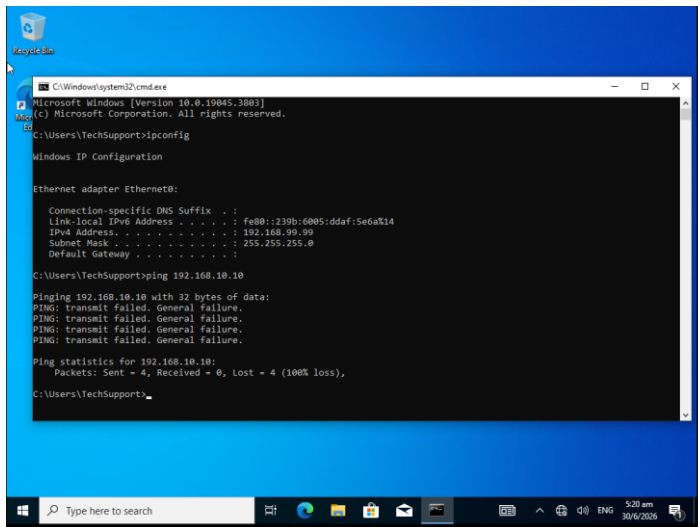
- **Core Server Node:** Windows Server 2022 Domain Controller (192.168.10.10)
- **Target Endpoint:** Windows 10 Pro Workstation
- **Diagnostic Terminal:** Windows Command Prompt (cmd)

3. Creating the Network Failure (Simulating the Issue)

To practice real-world diagnostics, the network settings on the Windows 10 workstation were manually broken to simulate an accidental static IP misconfiguration or configuration mismatch.

4. Confirming the Connection drop via Terminal

1. Attempt to test data packet delivery to the central server by running “ping” command.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TechSupport>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

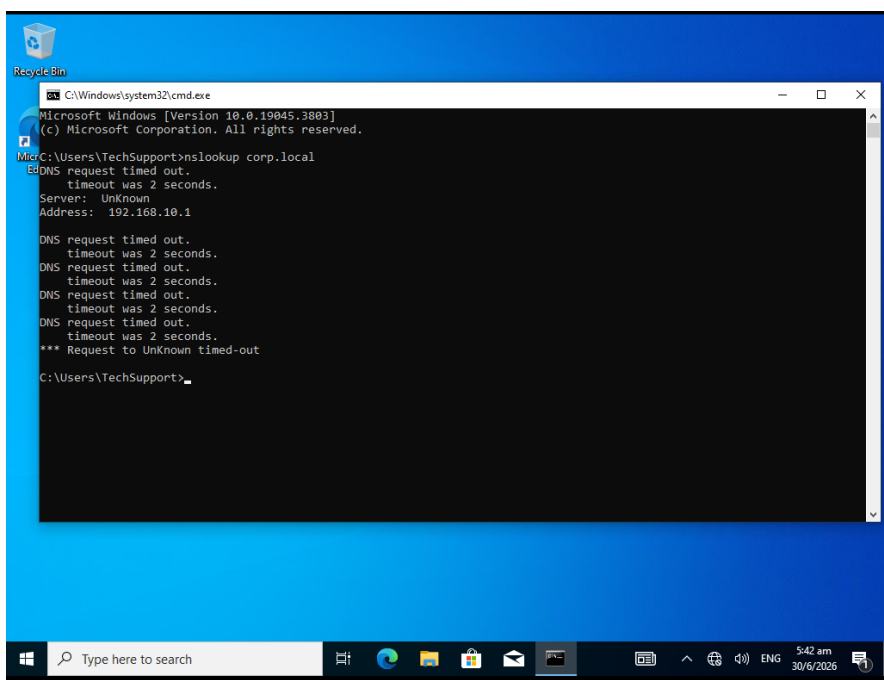
    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::230b:6905:ddaf:5e6a%14
    IPv4 Address. . . . . : 192.168.99.99
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\Users\TechSupport>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.
PING: transmit failed. General failure.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

2. Because the client workstation is on the wrong subnet, data packets cannot reach the server, resulting in a “**Transmit failed. General failure**”.
3. To track down exactly why the computer cannot communicate, run a **DNS lookup command** – “nslookup corp.local”. Test if the computer can translate server name into IP address.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TechSupport>nslookup corp.local
DNS request timed out.
    timeout was 2 seconds.
Server: Unknown
Address: 192.168.10.1

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to Unknown timed-out

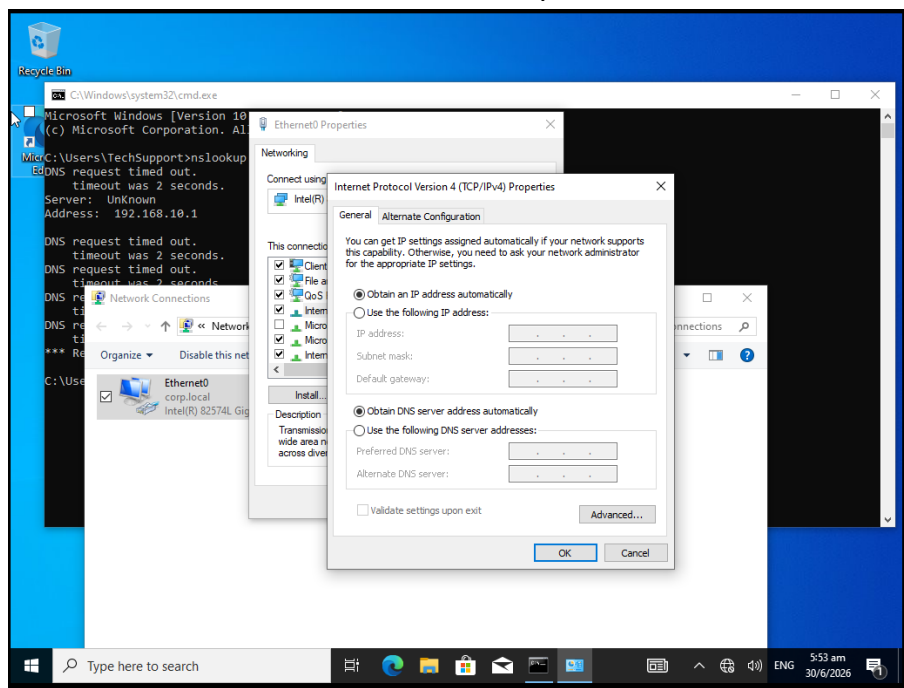
C:\Users\TechSupport>
```

4. This failure confirms a dual problem, **invalid IP address**, and **incorrect network DNS setting**.

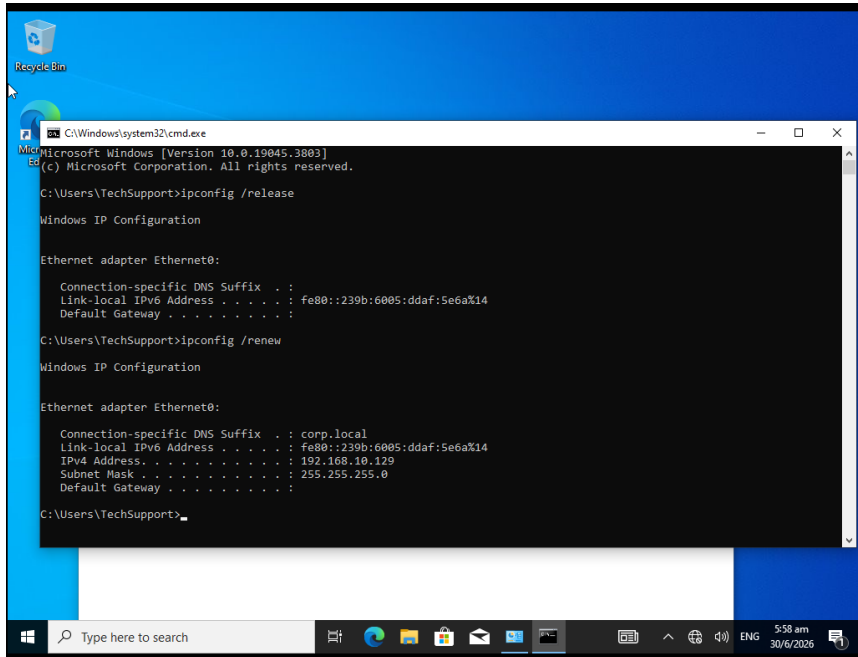
4. Repairing the Connection and Restoring Access

To resolve the issue, the client computer must return to automatic deployment so it can request a clean configuration from the server's active DHCP pool.

1. Open the run box (Win + R), type "ncpa.cpl" to bring up the **Network Connection** window.
2. Right-click on the **Ethernet**, and select **Properties**.
3. A Ethernet Properties window will pop-up.
4. Double-click Internet Protocol Version (TCP/IPv4), a IPv4 window will pop-up.
5. Change both the IP and DNS settings to "Obtain an IP address automatically" and "Obtain DNS server address automatically" and click OK.



6. Return back to the command prompt terminal and force the client machine to drop its old broken identity and request a fresh, verified network identity from the server.
7. Type “ipconfig /release” follow by “ipconfig /renew”.



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TechSupport>ipconfig /release

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::239b:6005:ddaf:5e6a%14
    Default Gateway . . . . . : 

C:\Users\TechSupport>ipconfig /renew

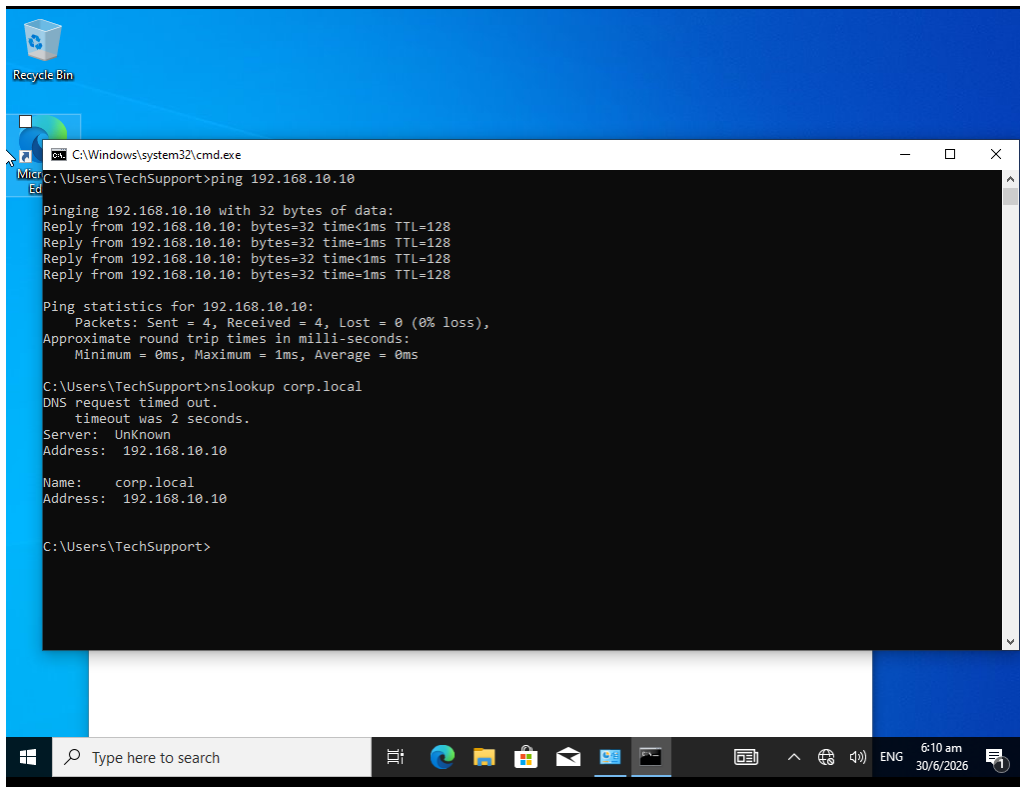
Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : corp.local
    Link-local IPv6 Address . . . . . : fe80::239b:6005:ddaf:5e6a%14
    IPv4 Address. . . . . : 192.168.10.129
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

C:\Users\TechSupport>
```

8. Verify the new setting by running ipconfig, it should show the new identity IP address.
9. Run final confirmation test: “ping 192.168.10.10” and “nslookup corp.local”.
10. The server now should response with active replies, translate corp.local to ip address, proving the connection is fully repaired.



```
C:\Windows\system32\cmd.exe
C:\Users\TechSupport>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\TechSupport>nslookup corp.local
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: 192.168.10.10

Name:   corp.local
Address: 192.168.10.10

C:\Users\TechSupport>
```

